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Evaluation of AI Chatbot Responses to Pediatric Urology Frequently Asked Questions

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A B S T R A C T

Objective: To evaluate the quality of responses from 4 publicly available LLMs (ChatGPT-4o, Claude 3.7, Gemini 2.5, and Copilot) to frequently asked questions (FAQs) in pediatric urology.

Methods: FAQs were generated using standardized prompts and submitted to each LLM using parent-centered instructions. Two board-certified pediatric urologists independently rated responses across 7 domains as follows: accuracy, completeness, safety, clarity, actionability, conciseness, and Global Quality Score using a 5-point Likert scale. Emotional tone was analyzed using the NRC Word-Emotion Association Lexicon, and readability was assessed using 7 established metrics.

Results: All LLMs produced generally high expert ratings for accuracy (mean 4.27), safety (4.36), and clarity (4.42). Claude achieved the highest overall quality score (4.35), followed by Gemini (4.03), ChatGPT (4.00), and Copilot (3.80). Emotional tone was predominantly positive and supportive across models. Reading corresponded to an 11th-15th grade reading level, exceeding the recommended 6th-8th grade patient-education standards.

Conclusion: LLMs can provide accurate, safe, and supportive information for parents, but their usefulness is limited by gaps in completeness and high readability levels. Claude achieved the highest overall quality, whereas ChatGPT demonstrated high safety with lower completeness. Future development would prioritize plain-language optimization, context-aware emotional framing, and parent co-design to improve comprehension.

Effective communication and access to reliable information are crucial in healthcare, particularly for the sensitive conditions managed by pediatric urologists. [1] Parents frequently seek rapid answers online before or after clinic visits; however, individual health literacy and emotional stress may affect their understanding. [1–3]

Large language model (LLM) chatbots are widely used artificial intelligence (AI) tools that can provide accessible health information to the public.[2,3] However, uncertainties remain regarding the accuracy, clarity, consistency, emotional tone, and safety of their responses. Unlike trained healthcare providers, LLMs may generate inconsistent or emotionally inappropriate responses, potentially compromising parental understanding and decision-making.[2,3]

Existing pediatric urology studies have reported up to 92% completeness in answering FAQs.[1] We aimed to utilize a standardized, parent-centered evaluation framework including traditional domains such as accuracy and readability,[4,5] along with emotional tone, an underexplored factor that may directly influence parental trust, anxiety, and healthcare-seeking behavior. We hypothesized that LLMs would provide generally accurate and safe responses to pediatric urology FAQs, but would differ in completeness, readability, emotional tone, and overall quality.

METHODS

We conducted a cross-sectional study to evaluate the accuracy, completeness, safety, readability, emotional tone, and overall quality of responses generated by publicly available AI chatbots to pediatric urology FAQs. Our 2 main objectives were to compare response quality across models and to generate a representative, parent-centered FAQ dataset reflective of real-world parental concerns. Four widely accessible LLMs were selected for their public availability, popularity, and widespread use, including ChatGPT-4o (OpenAI), Copilot (Microsoft), Gemini 2.5 (Google), and Claude 3.7 (Anthropic). All models were accessed through their publicly available interfaces using default settings. To avoid personalization effects and algorithmic memory bias, each LLM was accessed using a newly created email account and a new session.

As the study involved no human subjects or biological data, it was exempt from IRB review.

FAQ Generation and Response Collection

To generate a representative FAQ dataset, we designed 2 standardized prompts focused on common parental concerns in pediatric urology (Supplementary Table 1)

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Each LLM was prompted to generate 20 FAQs in early June 2025. The resulting questions from all models were combined into a single pool. Duplicate, overly general, repetitive, and highly condition-specific questions were removed to improve representativeness. We aimed to evaluate a manageable selection of common clinically relevant questions related to pediatric urology. (Supplementary Table 2)

Each question was submitted in a new session using a standardized prompt without follow-up questions to prevent context carryover. (Supplementary Table 1) The final dataset included 80 model responses to FAQs from June 2025. Full responses are provided in Supplementary Table 3.

Response Evaluation

Two board-certified pediatric urologists independently evaluated each LLM-generated response using a 5-point Likert scale for Global Quality Score (GQS) (1 = poor to 5 = excellent) and 6 specific domains as follows: accuracy, completeness, safety, clarity, actionability, and conciseness (1 = Strongly Disagree to 5 = Strongly Agree). Each domain was accompanied by a predefined, standardized definition provided to reviewers to ensure uniform scoring criteria and minimize subjective variation.

Emotional Tone

Emotional tone was analyzed using the NRC Word-Emotion Association Lexicon (version 0.92), which quantifies emotions by linking words to 8 emotion categories (anger, anticipation, disgust, fear, joy, sadness, surprise, and trust) and positive and negative sentiments based on human annotation.[6] For each response, we calculated the proportion of emotion-associated words (normalized by total word count) to allow comparison across responses of different lengths. Total emotional density was defined as the sum of all ten category densities for each response. Higher values indicate greater use of emotion-associated language.

Readability Assessment

To evaluate readability, we applied 7 established metrics. The Flesch Reading Ease score was used for overall text difficulty. It ranges from 0 to 100, with higher scores indicating easier readability; for example, scores between 60-70 generally reflect an 8th-9th grade, and 50-60 scores reflect a 10th-12th grade reading level.[7,8] The Gunning Fog Index and Flesch-Kincaid Grade Level estimate the education level required for comprehension, with a score of 12 corresponding to a 12th-grade level. The SMOG Index, commonly used in medical and health literacy contexts, evaluates syllable density to assess reading difficulty; a score of 12 indicates suitability for individuals with at least a 12th-grade reading level.[7,9] We also measured sentence length, word length, and syllables per word to validate readability scores.

Statistical Analysis

Statistical analyses were performed using SPSS version 29.0. Because all LLMs answered each question, repeated-measures comparisons of readability metrics, emotional tone, and expert ratings across models were conducted using nonparametric Friedman tests. Effect sizes for Friedman tests were summarized using Kendall's W. When overall Friedman tests were significant, post hoc pairwise comparisons were performed using Wilcoxon signed-rank tests with Holm correction (WSRT-Holm) for multiple comparisons. Inter-rater agreement between the 2 pediatric urologists was assessed separately for each domain using quadratic-weighted Cohen's kappa. Statistical significance was defined as $P < .05$.

RESULTS

Accuracy, Safety, and Relevance

Across all questions, the models demonstrated generally high performance. (Fig. 1) Claude was the top performer, achieving the highest overall GQS (4.35), followed by Gemini (4.03), ChatGPT (4.00), and Copilot (3.80). Friedman testing demonstrated significant between-model differences in GQS ($\chi^2 = 14.73$, $P = .002$), Actionability ($\chi^2 = 13.85$, $P = .003$), Completeness ($\chi^2 = 12.97$, $P = .004$), and Safety ($\chi^2 = 10.19$, $P = .01$). WSRT-Holm demonstrated significantly higher Completeness scores for Claude than ChatGPT (adjusted $P = .029$) and Copilot (adjusted $P = .003$), higher Actionability for Claude than Copilot (adjusted $P = .007$), and higher GQS for Claude than Copilot (adjusted $P = .002$). Although overall Safety differed significantly among LLMs, no pairwise comparison remained statistically significant after Holm correction. No significant between-model differences were observed for Accuracy, Clarity, or Conciseness.

Question-level analysis revealed a range of GQS, from 3.80 to 4.57. Q5 (mean 4.57), Q6 (4.52), and Q15 (4.32) had the highest average scores, while the lowest scores were recorded for Q16 (3.80), Q20 (3.98), and Q19 (4.07). Several questions revealed wide variability across LLMs (eg, Q10, Q20). A linear mixed-effects model demonstrated significant variation in GQS across questions (likelihood ratio test: LR = 30.90, $P = .041$), supporting non-random variation in question difficulty. Question-level results are provided in Supplementary Table 5.

Variability in GQS across questions differed by model (Brown–Forsythe $F(3, 76) = 4.00$, $P = .010$), with greater dispersion for Gemini and lower within-model variability for Claude and Copilot. Across the 80 responses, physician consultation was recommended in 72 (90%), either as a general recommendation (53 responses), when parental concerns persisted (6 responses), or when red-flag symptoms (eg, fever or swelling) were present (11 responses). Nine responses (11.3%) did not recommend physician consultation. Explicit medical-advice disclaimers were uncommon, occurring in only one Gemini response (1.3%). (Supplementary Table 4). These findings highlight inconsistencies in clinical guidance and safety-related messaging across models.

Inter-rater agreement varied across domains. Quadratic-weighted Cohen's kappa demonstrated fair agreement for Completeness ($\kappa w = 0.380$), but poor-to-slight agreement for Accuracy, Safety, Clarity, Actionability, Conciseness, and GQS (κw range, −0.013 to 0.120). Reviewer #1 systematically assigned higher scores than Reviewer #2, with mean differences ranging from 0.887 to 1.262 points across

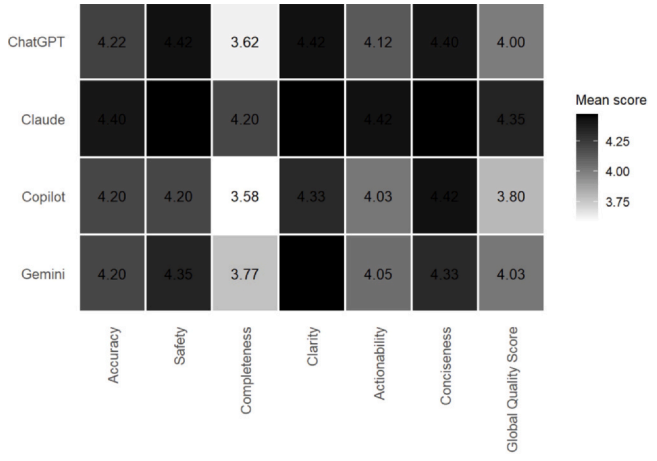


Figure 1. Average expert scores across evaluation domains for each LLM. Darker shading represents higher mean expert scores. LLM, large language model.

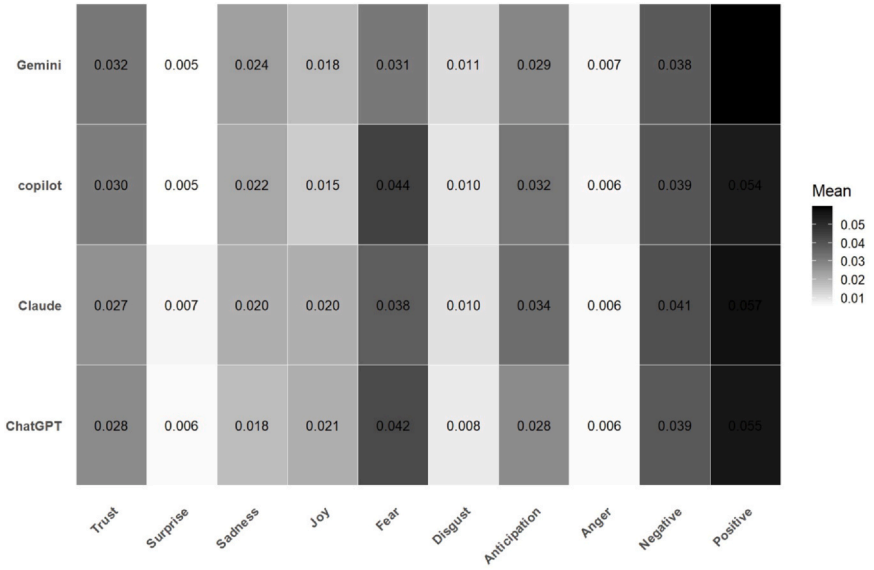


Figure 2. Emotional density across LLMs based on the NRC Emotion Lexicon. Darker shading indicates higher emotional density. LLM, large language model.

domains (all $P < .001$). Although the weighted kappa values were relatively low, both reviewers consistently rated the LLM responses favorably, and very few scores of 1-2 were assigned overall.

Emotional Analysis

A summary of emotion densities by question is provided in Supplementary Table 6, and the mean emotional density by LLM is shown in Figure 2.

Across all questions, Claude had the highest mean total emotional density (0.260), followed by Copilot (0.258), Gemini (0.251), and ChatGPT (0.251). However, total emotional density did not differ significantly between models (Friedman $\chi^2(3) = 0.30$, $P = .96$; Kendall's $W = 0.005$).

When analyzed by emotion category, densities differed significantly (Friedman $\chi^2(9) = 447.55$, $P < .001$; Kendall's $W = 0.622$). "Positive" language showed the highest mean density (0.0566), followed by "Negative" (0.0391), "Fear" (0.0390), and "Trust" (0.0291), whereas "Anger" (0.0063), "Surprise" (0.0057), and "Disgust" (0.0100) had the lowest values. Pairwise Wilcoxon testing demonstrated that the density of "Positive" language was higher than both "Fear" ($P = 4.35 \times 10^{-6}$) and "Trust" ($P = 1.03 \times 10^{-13}$).

Readability Assessment

Detailed readability scores are presented in Supplementary Table 6.

Flesch Reading Ease scores ranged from 38.6 (Claude) to 51.7 (ChatGPT), and SMOG and Flesch-Kincaid scores corresponded to an ≥11th-grade reading level across all models. Readability differed significantly across models ($P < .01$), driven by higher complexity in Claude; after excluding Claude, differences were not significant. (Table 1)

DISCUSSION

In this study, we evaluated the performance of 4 publicly available LLMs (ChatGPT-4o, Copilot, Gemini, and Claude) in responding to FAQs related to pediatric urology, focusing on accuracy, emotional tone, and readability. While LLMs are increasingly accessible for generating medically relevant content, their performance in specific clinical scenarios remains relatively underexplored. Our findings highlight both the potential and the limitations of AI-assisted resources for parents seeking information outside of clinical visits. Most models provided reasonably accurate, safe, and relevant information. However, important nuances emerged regarding emotional tone and readability, suggesting that the choice of model may influence the utility of LLM's outside the clinical setting.

Using structured prompts is important, as it can significantly improve LLM performance.[10]

Accuracy and safety: Overall strong performance with notable gaps

All LLMs in our study provided generally accurate and safe responses, consistent with prior studies showing ChatGPT's high reliability in urological FAQs. [1,2,10] However, we also identified clinically meaningful gaps that may limit the safe application of LLMs in real-world scenarios. Claude achieved the highest GQS, driven by strong completeness and clarity. ChatGPT produced consistently safe and concise responses but occasionally lacked completeness, limiting its clinical usefulness. Gemini's performance was the most variable, as it omitted key medical recommendations in several cases (eg, failing to advise a doctor visit in critical scenarios) and sometimes withheld complete guidance. Such omissions may delay care and have direct clinical consequences. Gemini's built-in safety mechanisms clearly restricted certain responses. Importantly, its failure to recommend medical consultation in specific scenarios raises concerns about its reliability as a safety-conscious language model. Copilot's answers were

Table 1
Overall readability scores.

LLM	Flesch Reading Ease	Flesch-Kincaid Grade	Gunning Fog Index	SMOG Index	Avg Sentence Length	Avg Word Length	Syllables per Word
ChatGPT	51.65262	10.97978	13.74715	12.63625	19.82451	4.463683	1.59646
Claude	38.6283	13.2977	16.29133	14.62499	21.84233	4.861588	1.726202
Copilot	51.49395	10.26459	13.15655	12.42011	16.85653	4.626513	1.633944
Gemini	50.15873	10.45265	13.23013	12.46891	16.86388	4.524527	1.649639
P-Value	.001	.000	.001	.001	.003	.000	.000

accurate and clear, but their lower completeness was reflected in the lowest GQS. Interestingly, the completeness of answers was not driven by conciseness, indicating that LLMs can provide brief yet comprehensive answers. Performance also varied significantly by question, indicating that LLM reliability is not uniform across clinical scenarios. Lower scores and greater variability were observed in questions requiring escalation decisions or nuanced counseling, suggesting that context-dependent and safety-sensitive topics remain more challenging for current models. Although the evaluated LLMs frequently recommended professional consultation—either generally, when parental concerns persisted, or when red-flag symptoms were present (72/80 responses, 90%)—explicit disclaimers were uncommon. Only one Gemini response stated that it could not provide medical advice, and no model consistently indicated that its responses should not replace professional medical advice. Overall, these results suggest that although LLMs can produce useful content, their performance may vary significantly across question types and prompting conditions. These differences among the models have also been reported in other studies. Alasker et al. evaluated the accuracy of ChatGPT-3.5, ChatGPT-4, and Google Bard in answering prostate cancer FAQs. While there was no significant difference, ChatGPT-3.5 showed superior performance in questions related to general knowledge, answering 88.9% of them correctly compared to 77.8% for ChatGPT-4% and 22.2% for Google Bard ($P = .018$). [8]

In other studies, using cardiothoracic anesthesia board-style questions, Claude outperformed ChatGPT-4, but both models performed poorly overall and struggled most with complex reasoning and visual interpretation, underscoring domain- and task-dependent variability. [11] Similarly, in an emergency medicine question pool, Copilot had the lowest overall error rate while Gemini had the highest, and error rates increased for specific topics (eg, trauma) and wording patterns (eg, questions containing “probability”), supporting prompt- and question-style sensitivity. [12] In cardiovascular pharmacology, comparative testing found Gemini performing worst on advanced items and scoring lower on short-answer evaluations. [13] Beyond exam-style tasks, patient-facing FAQ work in palliative care found significant differences across multiple chatbots (ChatGPT, Bard, Gemini, Copilot, Perplexity) in readability and quality. The modified DISCERN score for ChatGPT was significantly lower than the others, and the GQS score was highest for Gemini and lowest for Copilot. [3] Finally, when LLMs were evaluated for adherence to clinical practice guidelines for lumbosacral radicular pain, only partial guideline concordance was observed, with substantial between-model variation in “match rate,” with Perplexity having the highest and Claude showing the lowest performance.[14]

Our findings align with prior studies of LLMs responding to pediatric urology questions. [1,2,7,10,15] Across these studies, inaccuracies are uncommon, but gaps in completeness and clinical nuance persist, consistent with our observation of strong Accuracy/Safety and more variable Completeness and Actionability. [1,10] Evidence from other clinical domains similarly suggests that even broadly appropriate responses may require clarification or lack important details, reinforcing that LLMs are best positioned as adjunct tools rather than substitutes for professional counseling. [15] Despite low statistical agreement between our 2 rating experts, very few responses were rated poorly in any category, suggesting that these differences in scoring calibration were rarely a fundamentally different assessment of response quality.

Readability: A Barrier to Equitable Access

All models produced content requiring an approximately 11th-13th grade reading level, exceeding the NIH/AMA recommendation of 6th-8th grade for patient education.[3] Claude was the least readable (closer to a college level), while ChatGPT, Copilot, and Gemini still exceeded recommended thresholds. Although we did not evaluate parent comprehension directly, this discrepancy may represent a significant barrier for populations with lower health literacy. These findings echo prior work: Onder et al. reported that ChatGPT-4o’s responses on vesicoureteral reflux were difficult to understand without a college-level education (FRE, FKGL, SMOG, GFI, CLI scores of the text were 26 ± 12,15 ± 2.5, 16.3 ± 2, 18.8 ± 2.9, 15.3 ± 2.2, respectively).[7] Hanci et al. (compared the readability of responses from 5 AI chatbots (Bard, Copilot, Perplexity, ChatGPT, and Gemini) to 100 FAQs about palliative care. Their findings revealed that all chatbots exceeded the sixth-grade level across 7 indices, and there were significant differences among the chatbots (Bard was the most readable, Gemini the least).[3] Alasker et al. evaluated the readability of 3 LLMs to answer FAQs about prostate cancer. Google Bard demonstrated the highest readability (FRE 54.7; FKGL 10.2; $P < .001$), making it more suitable for general patient education. In contrast, ChatGPT-3.5 and −4 produced more complex texts.[8] Although complex language may enhance perceived credibility, it risks alienating or confusing parents with lower health literacy, especially in a socially sensitive pediatric urology context, underscoring the need for plain-language optimization in future systems.

Emotional Analysis

A distinguishing feature of our study is the use of the NRC Emotion Lexicon for quantitatively assessing the emotional tone of LLM-generated responses. Across models, responses were predominantly neutral to mildly positive in emotional tone. Claude exhibited the highest overall emotional tone (mean 0.095), while Copilot and Gemini scored lower. All models predominantly expressed “Positive” emotions, followed by "Fear" and "Trust," indicating a balance between reassurance and concern in their responses. "Anger," "Surprise," and "Disgust" had very low average densities, which is appropriate for supportive medical advice. While such analyses are novel in pediatric urology topics, Fatahi et al. assessed the emotional expression of ChatGPT-3.5 in comparison to human-generated responses to Stack Overflow questions (a popular online question-and-answer platform where programmers collectively post and answer technical coding questions). Their analysis showed that ChatGPT responses were more emotionally consistent and more positive than human responses.[16] Although not investigated directly, Claude and ChatGPT’s slightly greater emotional richness may affect perceived empathy and parental reassurance; however, our analysis cannot directly assess this effect. As such, further work is needed to clarify the importance of emotional richness in LLM responses.

In summary, LLMs demonstrate strong potential as adjunct tools for parental education in pediatric urology, providing generally accurate and supportive information. However, variability in completeness, inconsistent escalation guidance, and reading levels exceeding recommended standards highlight important limitations that may impact safe and equitable use. These findings suggest that while LLMs can support patient education, they should not be relied upon as standalone decision-making tools. Future efforts should focus on improving readability, ensuring consistent safety signaling, and incorporating parent-centered design to better align AI-generated information with real-world clinical needs. An important area for future research is the development and evaluation of specialty-specific LLMs that undergo training with input from pediatric urologists and parents.

LIMITATIONS

This study has several limitations. First, we were unable to incorporate parents' perspectives, who constitute the primary target audience for this work. Their feedback is essential to directly evaluate the usability, clarity, perceived empathy, and readability of AI-generated responses. Second, while we prioritized standardization by accessing all LLMs through new accounts and clean sessions to minimize algorithmic memory bias, this approach does not replicate real-world conditions where prior interactions, personalization, and follow-up questions may influence AI responses. Furthermore, because LLMs are updated frequently, the findings represent a snapshot of model performance at the time the responses were generated and may not fully reflect future versions of these systems. Additionally, emotional tone tools may overlook nuanced or context-specific emotional cues, especially in sensitive healthcare conversations. Although readability metrics provide objective measures of textual complexity, they do not

directly assess user comprehension or account for diverse levels of health literacy among parents. Finally, our study focused on English-language responses and may not generalize to other languages or cultures.

CONCLUSION

LLMs can deliver generally accurate, safe, and supportive information for families, but gaps in completeness and consistently high reading levels limit their reliability as stand-alone resources. Performance varied by model and next steps should prioritize plain-language outputs, context-appropriate tone, and clinician–AI collaboration (eg, escalation prompts/guardrails), ideally developed with parent co-design to ensure these tools augment professional guidance.

Ethics Declarations

This study is exempt from IRB review, as the study involved no human subjects or biological data.

Disclosures

The authors declare that they have no relevant financial interests.

CRediT Authorship Contribution Statement

Najva Mazhari: Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Andrew Freedman:** Writing – review & editing, Supervision, Investigation. **Nadine Friedrich:** Writing – review & editing. **Timothy Daskivich:** Writing – review & editing, Methodology, Conceptualization. **Paul Kokorowski:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of Generative AI and AI-assisted technologies in the writing process

Generative AI and AI-assisted technologies were NOT used in the preparation of this work.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.urology.2026.08.006.

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